EV Battery Course — Module 3: Lithium-Ion Battery Chemistry

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The cathode sets most of a cell's cost, energy, safety and life; the anode sets much of its charging speed. This module covers each major chemistry and ends with a comparison matrix for LFP, NMC, NCA and sodium-ion.

All figures below are typical ranges for commercial EV cells, not guaranteed specifications. Actual values depend on cell design (electrode thickness, format), test conditions (temperature, C-rate, depth of discharge) and the manufacturer.

1. Learning objectives

- Explain the crystal structure and working principle of each major cathode family.
- Explain graphite, silicon-graphite, lithium-metal and hard-carbon anodes.
- Describe the electrolyte, separator, current collectors, binders and conductive additives.
- Compare chemistries on voltage, energy, power, cost, life, safety, charging, manufacturing and recycling.
- Explain why LFP wins cost-sensitive segments and nickel-rich wins where pack energy density matters.

2. Fundamentals: three crystal families

Every cathode is a host crystal that lithium moves in and out of. There are three families:

Family	Structure	Lithium moves in	Examples
Layered oxide	Sheets of metal oxide with lithium between them	2 dimensions	LCO, NMC, NCA
Spinel	3D framework of channels	3 dimensions	LMO
Olivine	Phosphate framework with 1D tunnels	1 dimension	LFP, LMFP

Layered oxides store the most energy but release oxygen when overheated. Olivines hold oxygen in strong P–O bonds, so they are much more thermally stable but store less energy.

3. Cathode chemistries in detail

LFP — lithium iron phosphate ($LiFePO_4$)

- **Working principle:** Fe^{2+}/Fe^{3+} redox; lithium moves through 1D tunnels. Two-phase reaction gives a very flat voltage plateau.
- **Reaction (discharge):** $FePO_4 + Li^+ + e^- \rightarrow LiFePO_4$
- **Voltage:** 3.2–3.3 V nominal; operating about 2.5–3.65 V.
- **Energy:** about 160–205 Wh/kg at cell level in current EV cells.
- **Cost drivers:** lithium carbonate, iron and phosphate are cheap and abundant; no nickel or cobalt.
- **Life:** about 3,000–6,000+ cycles; good calendar life.
- **Safety:** thermal runaway onset roughly 250–270 °C; little oxygen release.
- **Charging:** good; poor conductivity is solved by nano-sizing and carbon coating.
- **Manufacturing:** mature, high-yield; China dominates material production.
- **Recycling:** low metal value makes recycling less profitable than NMC.
- **Best for:** mass-market cars, two- and three-wheelers, buses, stationary storage.

NMC — lithium nickel manganese cobalt oxide ($LiNi_xMn_yCo_zO_2$)

- **Working principle:** layered oxide. Nickel supplies most of the capacity (Ni^{2+}/Ni^{4+}), manganese stabilises the structure, cobalt keeps the layers ordered and improves rate capability.
- **Grades:** named by ratio: NMC111, 532, 622, 811 (80% Ni). Higher nickel means more energy, less stability, less cobalt.
- **Voltage:** 3.6–3.7 V nominal; operating about 2.8–4.2 V.
- **Energy:** about 220–300 Wh/kg at cell level (higher for nickel-rich grades).
- **Cost drivers:** nickel and cobalt prices, which are volatile.
- **Life:** about 1,000–2,500 cycles, depending on grade and use.
- **Safety:** runaway onset roughly 200–220 °C, lower for higher nickel; releases oxygen.
- **Charging:** good to very good.
- **Manufacturing:** high-nickel grades are moisture-sensitive and need tighter humidity control.
- **Recycling:** high value from Ni and Co; the recycling industry's economics depend on it.
- **Best for:** long-range and premium vehicles where pack energy density matters.

NCA — lithium nickel cobalt aluminium oxide ($LiNiCoAlO_2$)

- **Working principle:** layered, very nickel-rich (typically about 80–90% Ni); a small aluminium dopant stabilises the lattice.
- **Voltage:** about 3.6 V nominal; 2.7–4.2 V.
- **Energy:** about 250–300 Wh/kg.
- **Life:** about 1,000–2,000 cycles.
- **Safety:** narrow thermal window; needs strong cooling and a strict BMS.
- **Best for:** high-energy cylindrical cells, historically Panasonic cells for Tesla.

LCO — lithium cobalt oxide ($LiCoO_2$)

- The original Sony (1991) commercial cathode. About 3.7 V, high volumetric density.
- Only about half the lithium can be removed safely, cobalt is expensive, and thermal stability is poor.
- Used in phones and laptops; the Tesla Roadster (2008) used LCO-based 18650 laptop cells. Not used in modern EVs.

LMO — lithium manganese oxide ($LiMn_2O_4$)

- Spinel; about 3.8 V; cheap and safe with good power.
- Manganese dissolves into the electrolyte at high temperature, hurting life. Used in early Nissan Leaf packs (blended with NMC). Largely replaced.

LMFP — lithium manganese iron phosphate ($LiMn_xFe_{1-x}PO_4$)

- Adds manganese to LFP. Manganese redox occurs at about 4.1 V vs about 3.45 V for iron, raising average voltage and energy by roughly 10–20% over LFP while keeping olivine safety.
- Challenges: lower conductivity, two voltage plateaus complicate SOC estimation. Often blended with NMC.
- Status: entering commercial cells from Chinese makers.

Emerging high-manganese and cobalt-free cathodes

- **LNMO ($LiNi_{0.5}Mn_{1.5}O_4$):** high-voltage spinel at about 4.7 V; needs electrolytes stable at that voltage.
- **Lithium-manganese-rich (LMR):** layered, high capacity and cheap; suffers voltage fade. Several OEMs have announced development programmes.
- **Ultra-high-nickel / cobalt-free layered oxides:** remove cobalt entirely; stability is the challenge.

4. Anode technologies

Anode	Capacity (mAh/g)	Strength	Weakness	Status
Graphite (natural or synthetic)	~372 theoretical, ~340–360 practical	Cheap, stable, low voltage, long life	Limited capacity; lithium plating if charged too fast or cold	Universal today
Silicon-graphite blend (a few % to ~10% Si)	~400–600	More energy, faster charge	Silicon swells up to ~300%; cracks, SEI regrowth	Commercial in many premium cells
Silicon-dominant	1,000–3,000+	Very high energy	Severe swelling; needs nano-structuring	Pilot and early commercial
Lithium metal	~3,860	Highest possible	Dendrites, safety, needs solid or special electrolyte	Lab to pilot
Hard carbon (for sodium-ion)	~250–350	Holds sodium, which graphite cannot	Lower density, first-cycle loss	Commercial in Na-ion
LTO (lithium titanate)	~175	Extremely long life and fast charge	Low voltage (~2.4 V cell), low energy	Niche (buses, industrial)

5. Other cell components

- **Electrolyte:** $LiPF_6$ (about 1 mol/L) in a mix of cyclic (ethylene carbonate) and linear (dimethyl, diethyl, ethyl methyl carbonate) carbonates. Additives such as vinylene carbonate and FEC shape the SEI. Flammable; sensitive to water.
- **Separator:** microporous polyethylene and/or polypropylene, about 9–25 µm, often ceramic-coated. **Shutdown:** polyethylene melts at about 130 °C, closing its pores and stopping ion flow, a built-in safety fuse. Ceramic coating stops the film shrinking at higher temperatures.

- **Current collectors:** copper foil (about 6–10 μm) for the anode, because aluminium alloys with lithium at low potentials; aluminium foil (about 10–15 μm) for the cathode, because copper oxidises at high potentials.
- **Binders:** PVDF (dissolved in NMP solvent) for cathodes; water-based CMC/SBR for anodes.
- **Conductive additives:** carbon black, carbon nanotubes; a few percent of electrode mass to carry electrons through poorly conductive active material.

6. Step-by-step: why nickel-rich gives more energy but less safety

1. More nickel means more Ni^{2+} available to oxidise to Ni^{4+}, so more lithium can be removed per gram: more mAh/g.
2. In a highly charged state, Ni^{4+} is very unstable and readily reduced.
3. If the cell heats, the layered structure transforms and releases oxygen.
4. The oxygen burns the electrolyte inside the cell. That is why thermal runaway in nickel-rich cells is violent.
5. LFP's oxygen is locked in phosphate groups; no oxygen release, so a failing cell smoulders and vents rather than igniting fiercely.

7. Engineering equations

Theoretical capacity (from Module 2):

$$Q_{th} = \frac{n F}{3.6 M}\ \ (\text{mAh/g})$$

Specific energy of an active material:

$$e_{material}\ (\text{Wh/g}) = \frac{Q\ (\text{mAh/g}) \times V_{avg}}{1000}$$

Mass of active material per kWh:

$$m\ (\text{kg/kWh}) = \frac{1}{e_{material}\ (\text{Wh/g})}$$

Metal content per kWh:

$$m_{metal} = m_{material} \times \frac{x \, M_{metal}}{M_{material}}$$

8. Worked examples

Example 1: NMC811 theoretical capacity. $LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$: M = 6.94 + 0.8(58.69) + 0.1(54.94) + 0.1(58.93) + 32.00 = **97.28 g/mol**.

- Q_th = 96,485 ÷ (3.6 × 97.28) = **275.5 mAh/g**. Practical about 200 mAh/g.

Example 2: cathode material per kWh.

Cathode	Practical capacity	Avg voltage	Wh/g	kg active material per kWh
LFP	155 mAh/g	3.2 V	0.496	**2.02**
NMC811	200 mAh/g	3.7 V	0.740	**1.35**

LFP needs about 50% more cathode mass per kWh (ignoring the anode and the cell hardware).

Example 3: nickel and cobalt per kWh in NMC811.

- Ni fraction = 46.95 ÷ 97.28 = 0.483 → 1.35 × 0.483 = **0.65 kg Ni/kWh**
- Co fraction = 5.89 ÷ 97.28 = 0.061 → 1.35 × 0.061 = **0.08 kg Co/kWh**
- For a 60 kWh pack: about **39 kg nickel and 5 kg cobalt** in the cathode.

Example 4: cell mass for a 60 kWh pack (cell level only).

- LFP at 170 Wh/kg: 60,000 ÷ 170 = **353 kg**
- NMC at 260 Wh/kg: 60,000 ÷ 260 = **231 kg**
- Difference about 122 kg before pack structure. Cell-to-pack designs narrow the pack-level gap, because LFP's safety allows less protective structure.

Example 5: illustrative cost comparison. Assume cell prices of $65/kWh (LFP) and $85/kWh (NMC). These are assumptions for the exercise, not quotes; real prices vary by year, region and volume.

- LFP 60 kWh: 60 × 65 = **$3,900**. NMC 60 kWh: 60 × 85 = **$5,100**. Difference $1,200 per vehicle at cell level.

9. Comparison matrix

Attribute	LFP	NMC (811-class)	NCA	Sodium-ion
Nominal cell voltage	3.2 V	3.6–3.7 V	3.6 V	~3.0–3.1 V
Cell specific energy	160–205 Wh/kg	240–300 Wh/kg	250–300 Wh/kg	120–175 Wh/kg

Attribute	LFP	NMC (811-class)	NCA	Sodium-ion
Power capability	Good	Very good	Very good	Good; strong at low temperature
Cycle life	3,000–6,000+	1,000–2,500	1,000–2,000	2,000–4,000+ (claimed)
Thermal stability	Highest	Moderate	Lowest	High
Cold performance	Weak	Moderate	Moderate	Strong
Critical minerals	Li only	Li, Ni, Co, Mn	Li, Ni, Co	None of Li, Ni, Co
Relative cell cost	Low	High	High	Potentially lowest at scale
SOC estimation	Hard (flat curve)	Easy	Easy	Moderate (sloping)
Recycling value	Low	High	High	Low
Maturity	Mass production	Mass production	Mass production	Early commercial
Typical use	Mass-market cars, 2W/3W, buses, storage	Long-range, premium	High-energy cylindrical	Budget city cars, 2W, storage

How conditions move these numbers: cycle life roughly halves for every ~10 °C of sustained operating temperature above about 25–30 °C (a rule of thumb, not a law); deeper depth of discharge, higher end-of-charge voltage and higher C-rates all shorten life. A power-optimised cell of any chemistry has lower Wh/kg than an energy-optimised one.

10. Real-world EV examples

- **LFP:** BYD Blade, Tesla standard-range models, Tata's Nexon and Curvv EVs, and Mahindra's BE 6 and XEV 9e (as reported at launch; check current specifications per variant). Most Indian electric two- and three-wheelers have moved to LFP.
- **NMC:** Hyundai Ioniq 5, VW ID family, most Korean and European long-range EVs.

- **NCA:** Panasonic cylindrical cells in Tesla vehicles.
- **Sodium-ion:** CATL and BYD have announced and begun supplying sodium-ion cells; deployment in vehicles is early and mainly in China.

11. Manufacturing applications

- **Humidity:** high-nickel NMC/NCA powders react with moisture and CO_2 to form surface lithium compounds; they need lower dew points than LFP lines.
- **Line flexibility:** cathode chemistry changes slurry recipes and drying, but most cell assembly equipment is shared. Sodium-ion can reuse many LFP lines, which helps its scale-up.
- **Supply risk:** NMC exposes you to Ni/Co price swings and sourcing rules; LFP exposes you to Chinese dominance in LFP cathode production.
- **Costing:** model cathode cost from metal content per kWh (Example 3) × metal price + processing. This is a should-cost lever you can use directly.

12. Common misconceptions

- **"LFP is old technology."** LFP cell energy has risen sharply through cell design and cell-to-pack integration.
- **"Cobalt-free means nickel-free."** Many cobalt-free cathodes are very nickel-rich.
- **"Higher Wh/kg always means more range."** Pack-level density, efficiency and usable window matter as much.
- **"Sodium-ion is a drop-in replacement."** It has lower energy, different voltage windows and needs BMS changes.
- **"Silicon anodes are all-or-nothing."** Most commercial cells add only a few percent of silicon.

13. Practical case study: switching a two-wheeler from NMC to LFP

Problem. An Indian scooter maker using NMC pouch cells faces warranty claims for heat-related degradation and a high-profile thermal incident; cost pressure is also high.

Analysis. NMC's lower thermal margin plus high ambient temperatures, minimal pack cooling and frequent full charges cause fast ageing. LFP offers higher thermal stability and longer life at lower cost, at the price of about 25–35% more cell mass for the same energy.

Solution. Switch to LFP prismatic or cylindrical cells; redesign the pack for the larger volume; recalibrate BMS SOC estimation for the flat voltage curve; adjust charger voltage limits (3.65 V/cell vs 4.2 V/cell).

Lesson. Chemistry is a system choice involving pack, BMS, charger and warranty strategy, not just a cell swap.

14. Revision notes

- Layered (NMC, NCA, LCO) = high energy, oxygen release risk. Olivine (LFP, LMFP) = safe, long-lived, lower energy. Spinel (LMO) = cheap, good power, short life.
- More nickel = more energy, less stability.
- Graphite dominates anodes; silicon is being blended in; lithium metal is next-gen.
- Copper for the anode collector, aluminium for the cathode.
- PE separator shuts down at about 130 °C.

15. Glossary

Term	Meaning
NMC811	80% Ni, 10% Mn, 10% Co by metal ratio
Olivine	LFP's crystal structure, with 1D lithium tunnels
Plateau	Flat region of a voltage curve
Hard carbon	Disordered carbon anode used for sodium-ion
PVDF / NMP	Cathode binder and its solvent
CMC/SBR	Water-based anode binders
Shutdown separator	Separator whose pores close when overheated
LMFP	LFP with manganese added for higher voltage

16. Knowledge check

Conceptual

1. Why does LFP not release oxygen when overheated while NMC does?
2. Why is copper used for the anode current collector and aluminium for the cathode?
3. Why does silicon improve energy but threaten cycle life?

4. Why is SOC estimation harder for LFP?
5. What does LMFP gain over LFP, and what does it give up?

Numerical

1. A 75 kWh pack uses NMC811 at 1.35 kg cathode per kWh. How much nickel does it contain?
2. Compare cell mass for a 40 kWh pack: LFP at 180 Wh/kg vs sodium-ion at 150 Wh/kg.
3. A sodium-ion cathode delivers 120 mAh/g at 3.1 V. How many kg are needed per kWh?

Application

1. Recommend a chemistry for: (a) an e-rickshaw fleet in Lucknow; (b) a 500 km premium SUV; (c) a home storage unit. Justify each.
2. Nickel prices double. How would you re-evaluate your should-cost model, and what design responses might an OEM consider?

17. Answer key

Conceptual

1. In LFP, oxygen is bound covalently in PO_4 groups with strong P–O bonds. In charged layered oxides, unstable Ni^{4+} and the layered lattice transform on heating and release lattice oxygen.
2. Aluminium alloys with lithium at the anode's low potential (about 0.1 V vs Li); copper oxidises and dissolves at the cathode's high potential (above about 3.5 V vs Li).
3. Silicon stores roughly 10× graphite's lithium per gram but expands up to about 300%, cracking particles and constantly exposing fresh surface where SEI grows, consuming lithium.
4. Its voltage is nearly flat between about 10% and 90% SOC, so voltage gives little SOC information.
5. It gains higher average voltage (Mn at about 4.1 V) and about 10–20% more energy while staying olivine-safe; it gives up conductivity and has two plateaus that complicate SOC estimation.

Numerical

1. 75 × 1.35 = 101.25 kg cathode × 0.483 = **48.9 kg nickel**.
2. LFP: 40,000 ÷ 180 = **222 kg**. Na-ion: 40,000 ÷ 150 = **267 kg**. Na-ion is about 44 kg heavier.
3. 120 × 3.1 ÷ 1000 = 0.372 Wh/g → 1 ÷ 0.372 = **2.69 kg/kWh**.

Application

1. (a) LFP, or sodium-ion as it matures: low cost, long cycle life, heat tolerance; mass less critical. (b) Nickel-rich NMC with silicon-graphite: highest pack energy density to fit 500 km in a practical pack. (c) LFP: cycle life, safety in a home, cost; mass irrelevant.
2. Update metal price inputs; recalculate cathode cost per kWh from Ni content (0.65 kg/kWh for 811); test sensitivity. Responses: shift lower trims to LFP or LMFP; negotiate index-linked pricing; consider manganese-rich chemistries; secure long-term nickel offtake.

18. Further learning

- *Lithium-Ion Batteries: Basics and Applications*, Korthauer (Springer).
- J. B. Goodenough & K.-S. Park, "The Li-Ion Rechargeable Battery: A Perspective," *JACS* 2013.
- A. Manthiram, "A reflection on lithium-ion battery cathode chemistry," *Nature Communications* 2020.
- Battery University BU-205, "Types of Lithium-ion."
- IEA *Global EV Outlook* (annual): chemistry market shares.

Next: Module 4, the evolution of EV battery technology.